

4.3.13 This figure can be interpreted as:
 Consider an $\triangle ABC$, where F, E are points on AB, AC such that B, C, F and E are concyclic. The angle bisector of $\angle BXC$ meets FE at D . Suppose $FB + CE = FE$. Prove that B, C, D and X are concyclic, where $FX = FB$.

Let I be the incentre of $\triangle ABC$, since $\angle BXC = 180^\circ - \angle BXC - \angle CXE$
 and $\angle BXC = 90^\circ - \frac{1}{2}\angle AFE = 90^\circ - \frac{1}{2}\angle ACB$ and $\angle CXE = 90^\circ - \frac{1}{2}\angle ABC$
 $\therefore \angle BXC = 90^\circ - \frac{A}{2} = 180^\circ - \angle BIC$

$\therefore B, X, I$ and C are concyclic.

Now $\angle BCX = 180^\circ - C - (90^\circ - \frac{B}{2}) = 90^\circ + \frac{B}{2} - C$ $\angle CBX = 180^\circ - (90^\circ + \frac{B}{2} - C + 90^\circ - \frac{A}{2})$
 $= C + \frac{A}{2} - \frac{B}{2} \Rightarrow \angle IBX = \angle IBC + \angle CBX = C + \frac{A}{2}$

Meanwhile $\angle IDX = 180^\circ - \frac{A}{2} - \angle AEF = C + \frac{A}{2} \Rightarrow I, B, D$ and X are concyclic.

Therefore, I, B, D, X and C are concyclic.

